



SKG14 IC8 Pacing and Emerging Technologies for ICM Engineers, L2

16.0– What is AI

July 2025

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Learning Outcome

At the end of the session, you will be able to:

- Explain the basic fundamentals of data statistics and analytics.
- Describe the common terminologies of AI and its subsects, ML and DL.
- Apply data analytics, AI and their subsets usage in various ICM applications.
- Describe the characteristics of computing requirements most suitable for AI.

16.0 What is AI?

- Introduction of AI
- Key Aspects & Goals of AI
- Branches of AI
- 5 Types of Data Analytics
- 4 Pillars of AI Analytics
- Evolution of AI Technique: Rule-based, ML and DL
- Standard Related to AI/ML

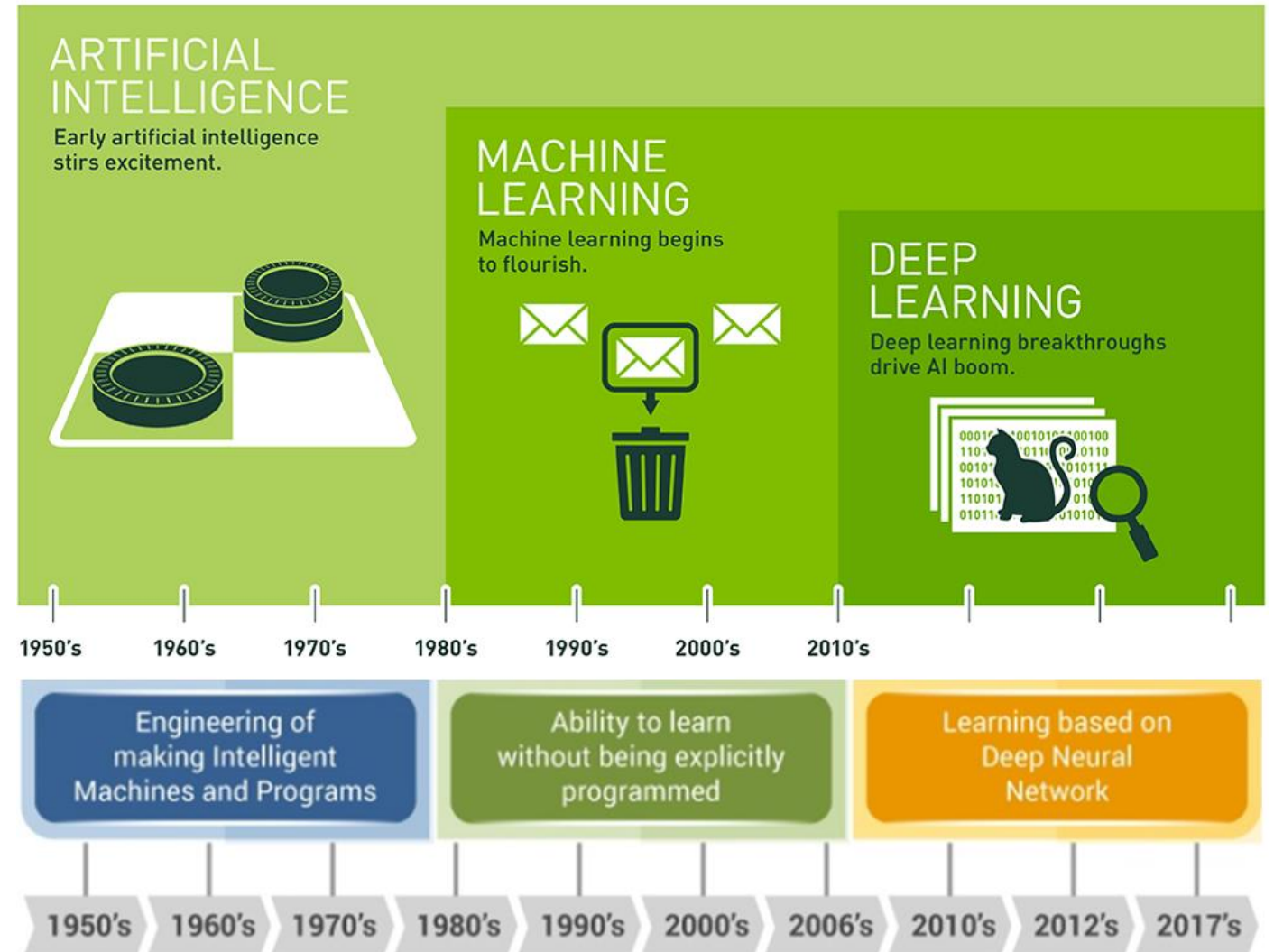
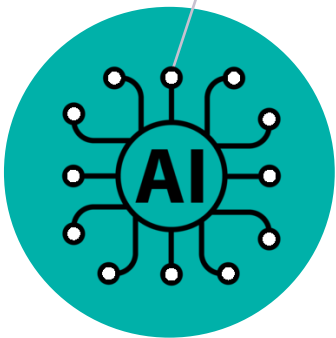
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Introduction of AI

According to ISO/IEC 22989:2022, Artificial Intelligence (AI) is defined as the "research and development of mechanisms and applications of AI systems."

An AI system is further described as an "engineered system that generates outputs such as content, forecasts, recommendations or decisions for a given set of human-defined objectives."

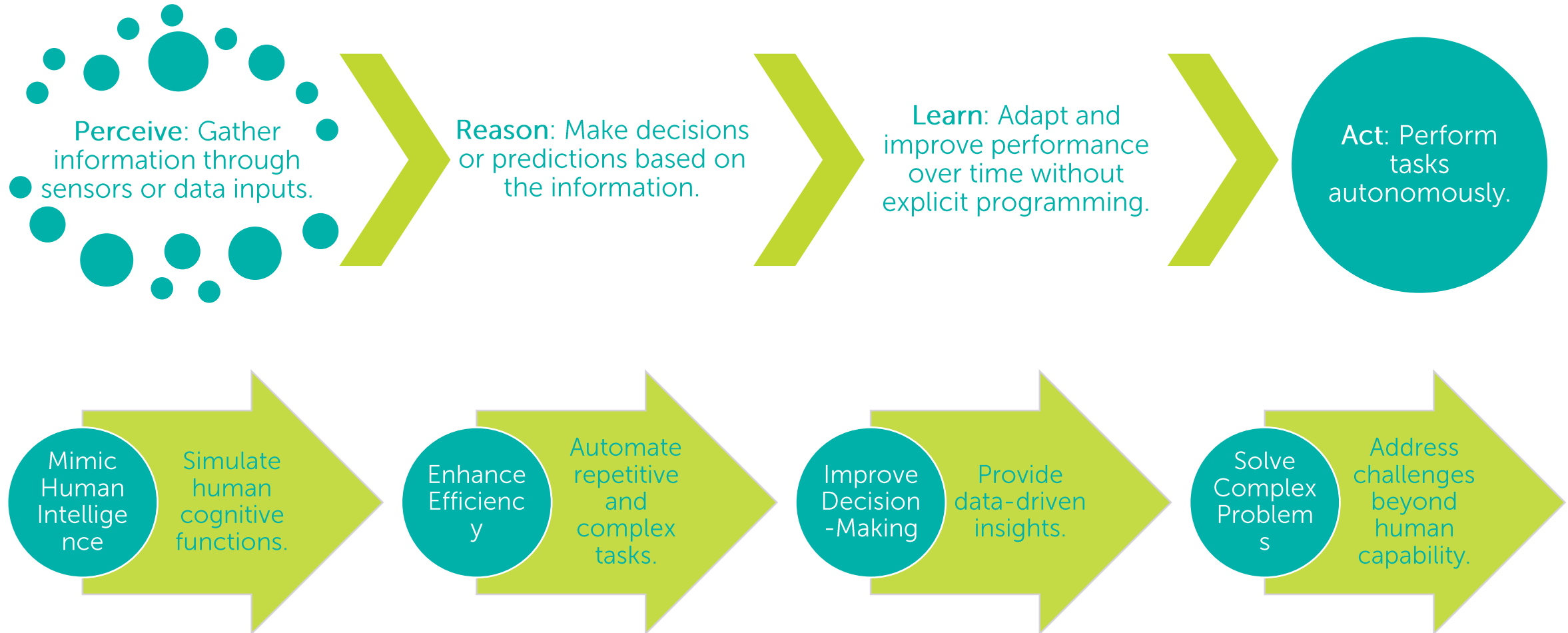


Source: https://doi.org/10.1007/978-3-319-94878-2_19

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Key Aspects & Goals of AI



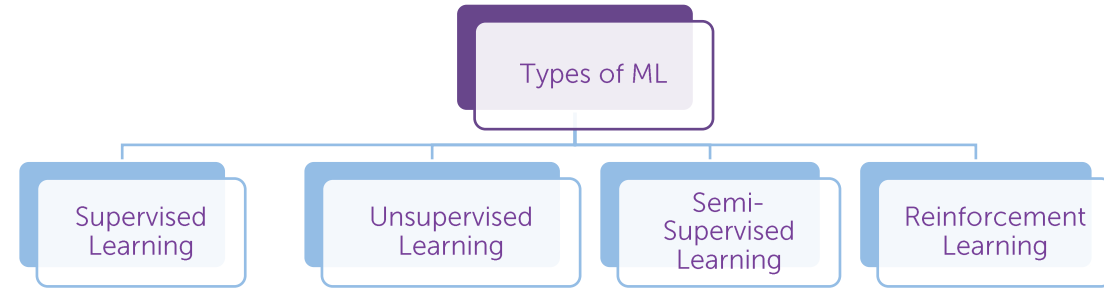
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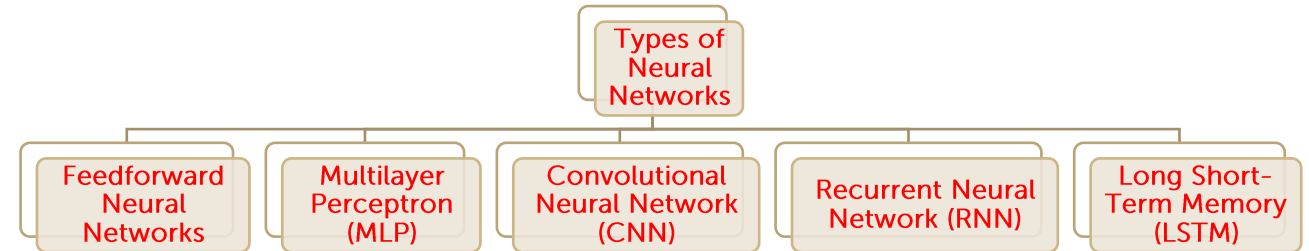
Branches of AI

Artificial Intelligence (AI) - AI is the science and engineering of making intelligent machines, especially intelligent computer programs.

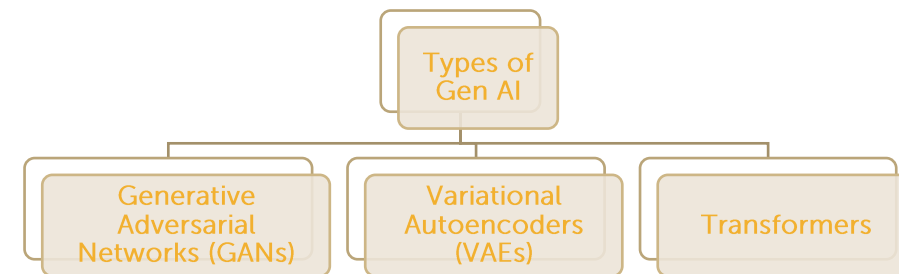
Machine Learning (ML) - ML is a subset of AI focused on developing systems that can learn and adapt from data without explicit programming.



Deep Learning (DL) - DL is a subset of ML that uses Artificial Neural Networks (ANN) with multiple layers to model and understand complex patterns in large datasets.

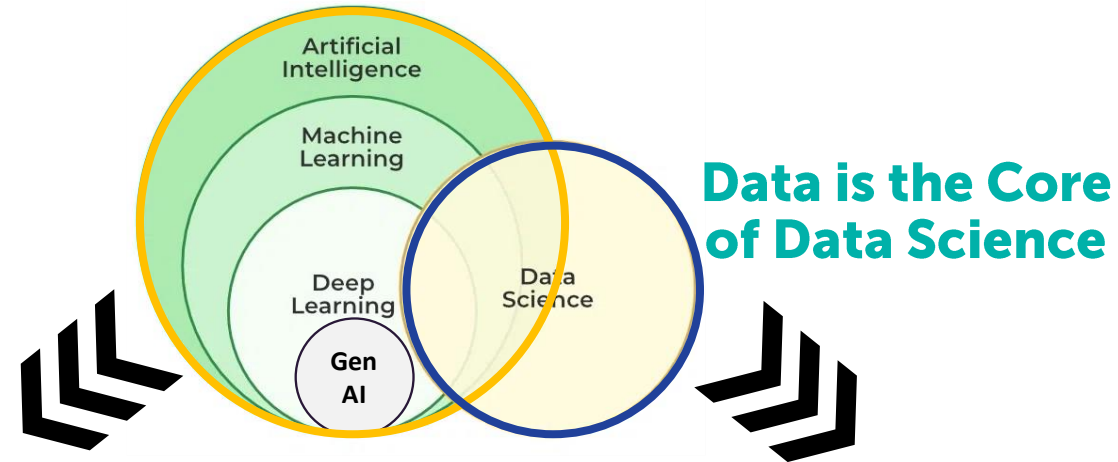


Generative AI (Gen AI) - Gen AI is a subset of DL that generates new content, such as text, images, music, code, and even videos, based on patterns learned from existing data.



Branches of AI

Data-Driven Insights is Enabled by Application of AI Techniques



Data is the Core of Data Science

AI Techniques



Machine learning

Is at the core of AI analytics, enabling models to learn from historical data and make predictions about future outcomes.



Natural language processing (NLP)

It is essential for analyzing and interpreting human language, enabling AI tools to process and understand text-based data.



Neural networks

They represent a series of algorithms that recognize relationships in data and are particularly useful in identifying complex patterns and correlations.

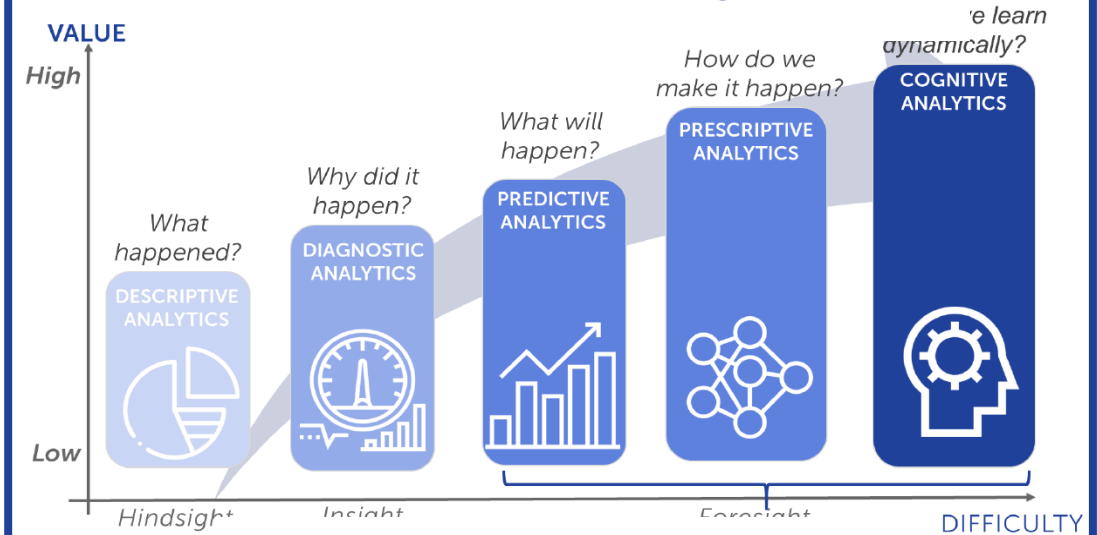


Deep learning

Implies processing vast amounts of unstructured data, such as images, videos, and text, to generate highly detailed and nuanced insights.

4 Pillars of AI Analytics

Data-Driven Insights



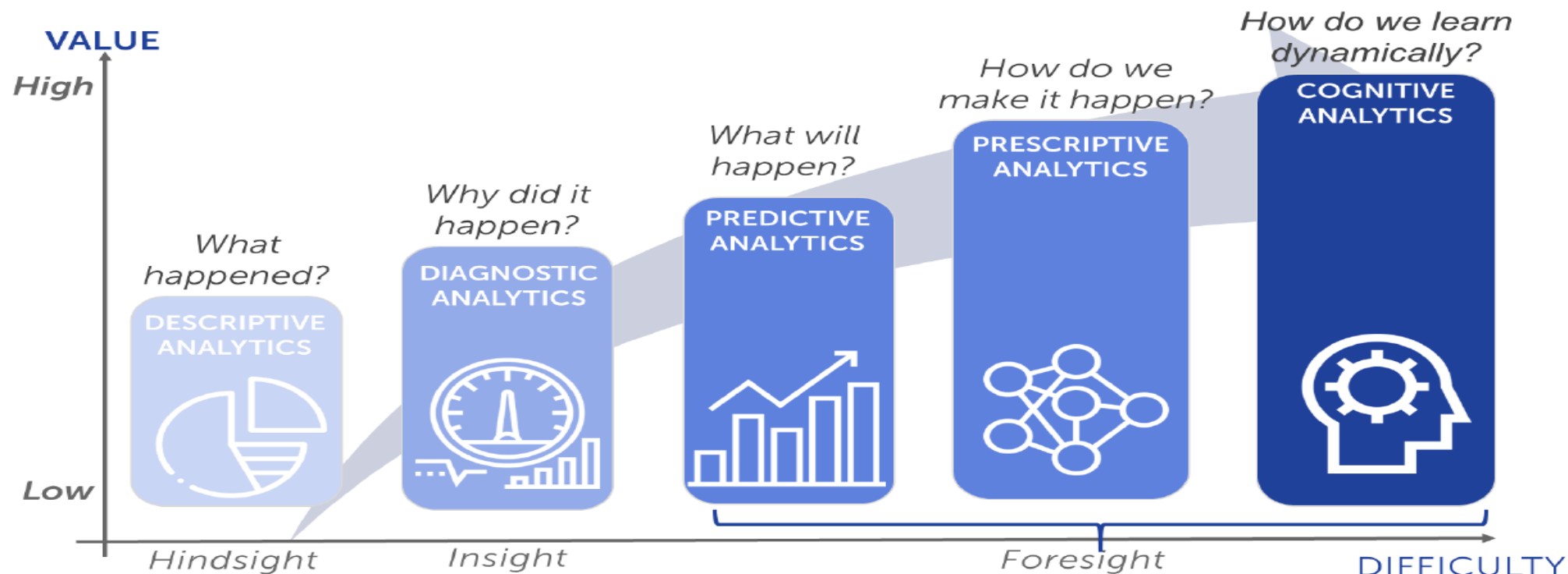
5 Types of Data Analytics

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5 Types of Data Analytics

Pillars varying levels of difficulty that is proportional to the potential value creation

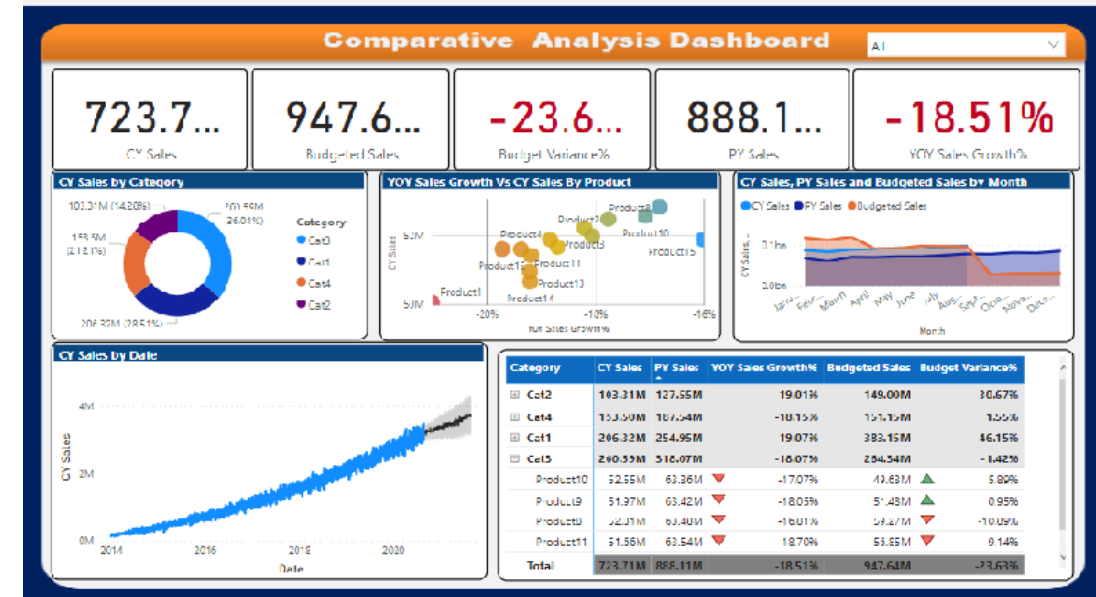


Source: PETRONAS Citizen Analytics Training Material

- Summarize and describe the characteristics of a dataset.
- May include calculating basic statistics such as mean, median, and mode or creating visualizations such as histograms or box plots.
- Identifies the cause of a problem or issue.
- May involve using data to diagnose a machine malfunction, identify the source of a customer complaint, or diagnose a medical condition.
- Involves using data to predict future events or outcomes.
- May include using AI/ML algorithms to build predictive models based on historical data.
- Provides recommendations or solutions based on data analysis.
- May involve using optimization algorithms to identify the best course of action or using decision trees to guide decision-making.
- leverages AI/ML to mimic human reasoning and decision-making.
- It acts as a forward-thinker, exploring "what could be possible," enabling to foresee trends, adapt dynamically, and drive innovation across industries.

5 Types of Data Analytics

Descriptive Analytics – Understand what has happened in the past



Source: [Microsoft Power BI](#)

Purpose

Understand what has happened in the past.

Focus

Summarizes historical data to provide insights into trends, patterns, and anomalies.

Examples

Sales reports showing revenue trends over the last year.

Website analytics dashboards displaying the number of daily visitors.

Techniques

Data visualization (e.g., charts, graphs).

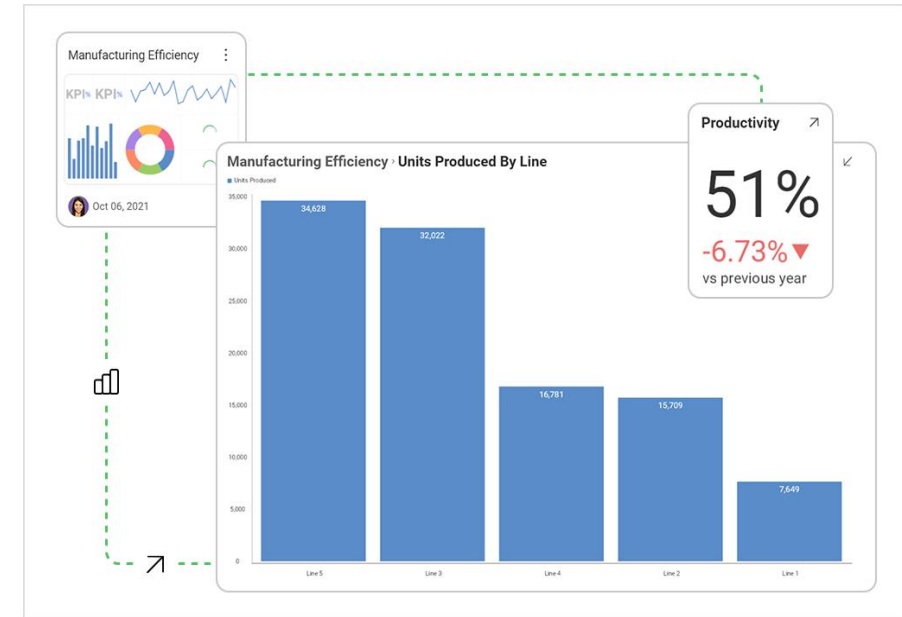
Summary statistics (e.g., mean, median, mode).

Tools

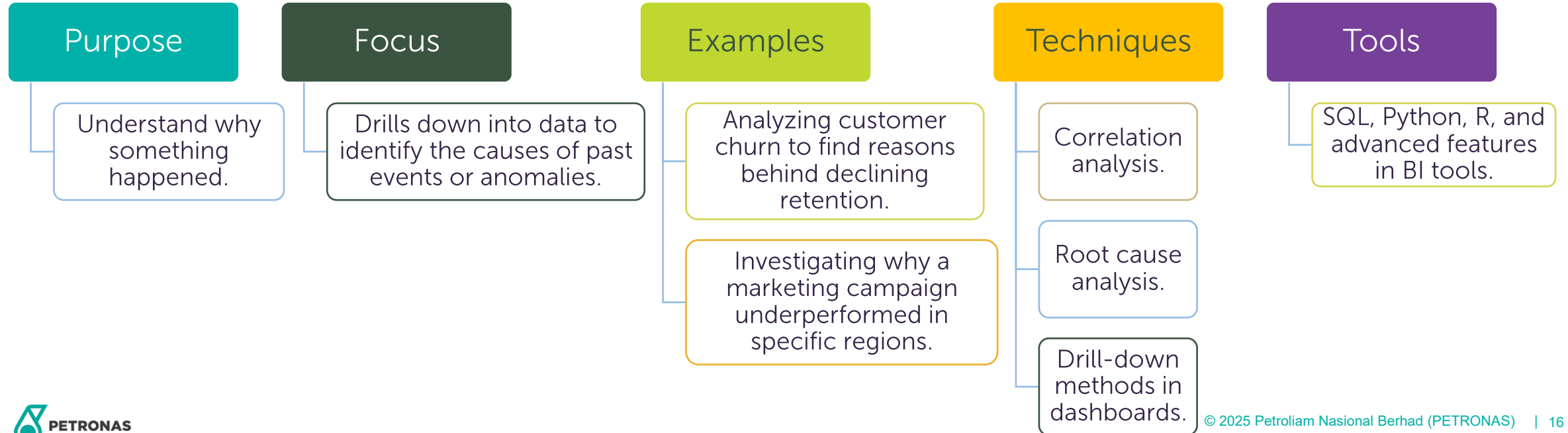
Excel, Tableau, Power BI, etc.

5 Types of Data Analytics

Diagnostic Analytics – Understand why something happened.

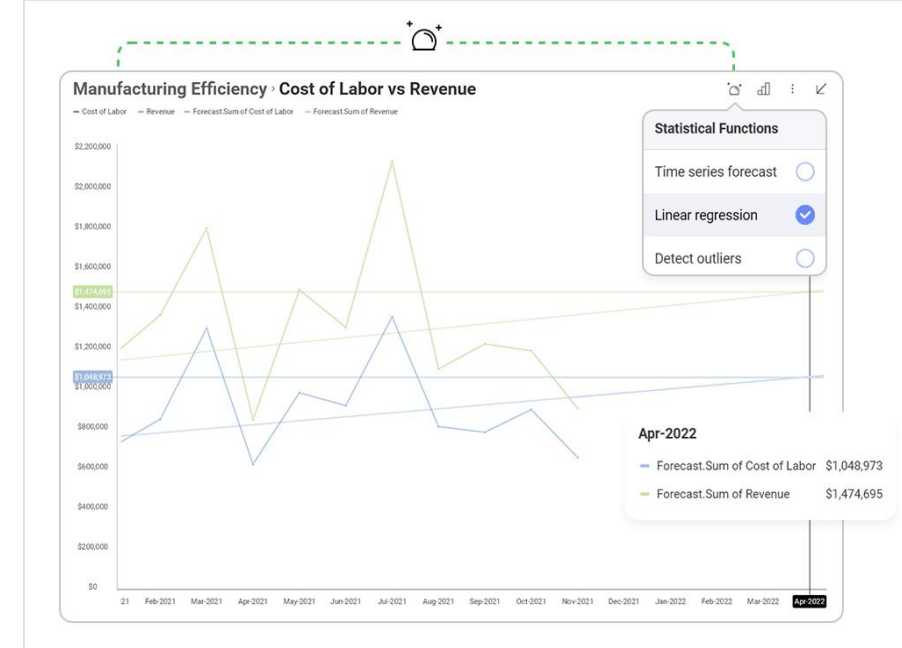


Source: <https://www.slingshotapp.io/blog/types-of-data-analysis>

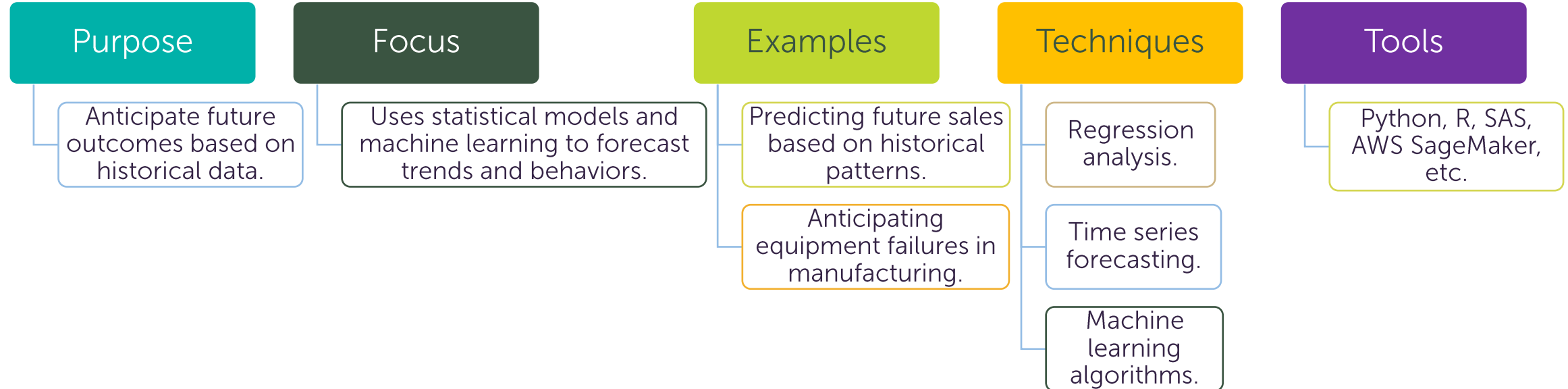


5 Types of Data Analytics

Predictive Analytics – Anticipate future outcomes based on historical data.

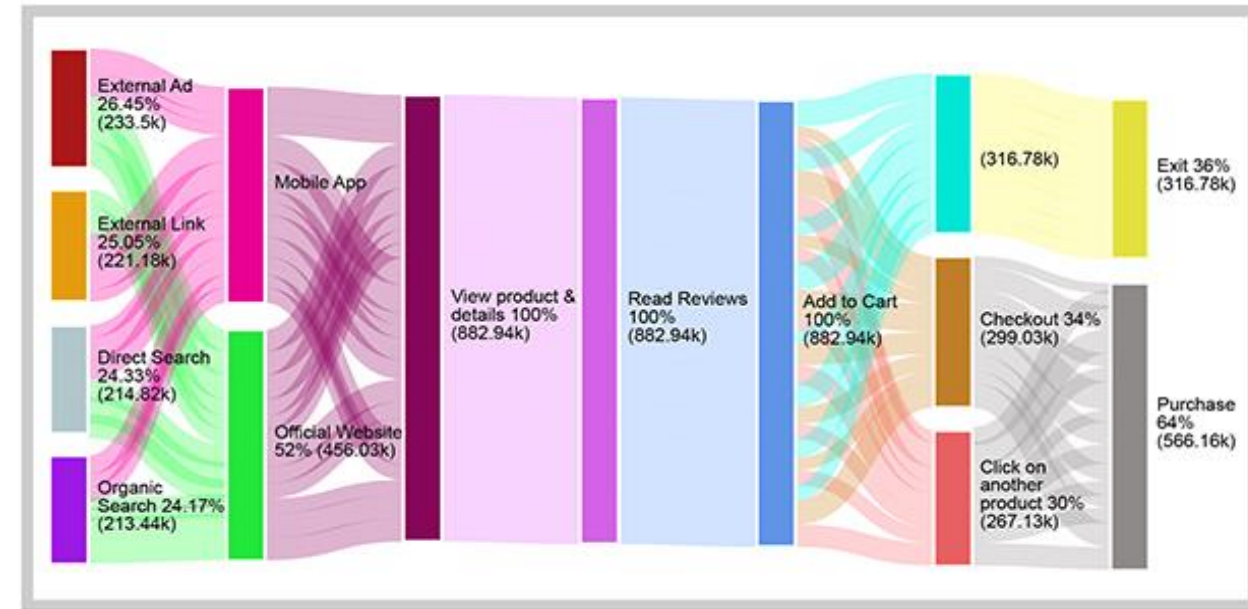


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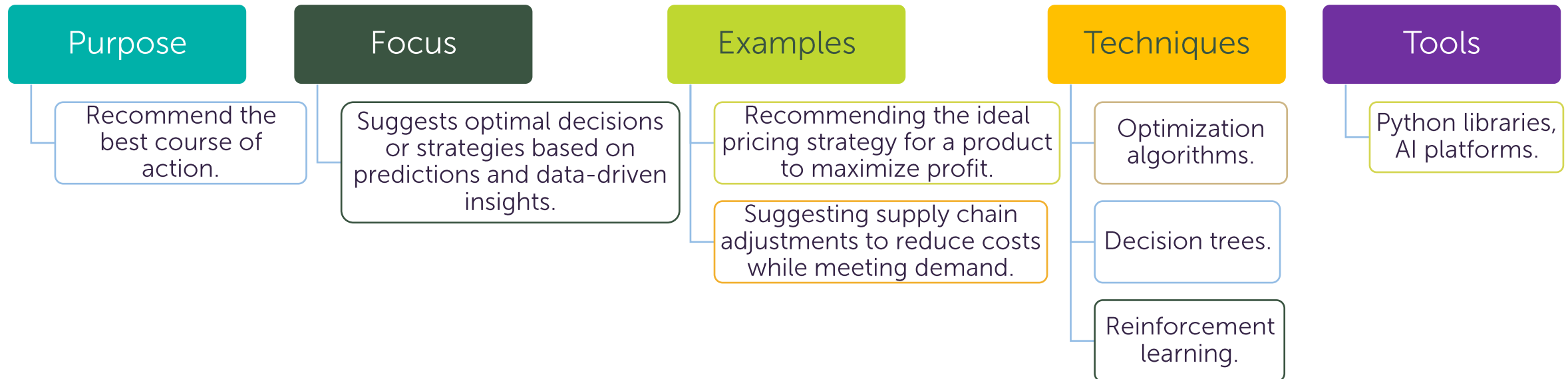


5 Types of Data Analytics

Prescriptive Analytics – Recommend the best course of action



Source: <https://www.slingshotapp.io/blog/types-of-data-analysis>

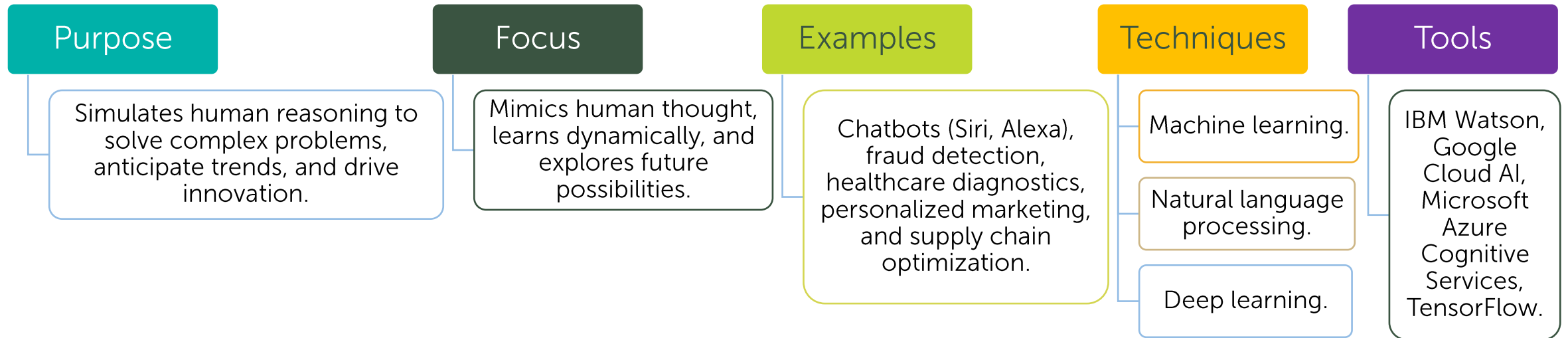


5 Types of Data Analytics

Cognitive Analytics – Simulate human reasoning and decision-making processes by combining data-driven insights with contextual understanding



Source: <https://www.slingshotapp.io/blog/types-of-data-analysis>

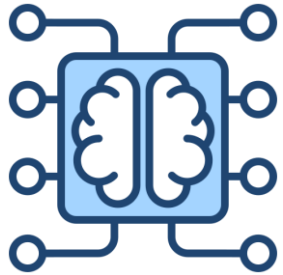


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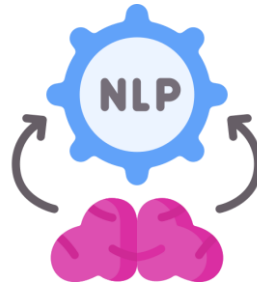
4 Pillars of AI Analytics

Technique Selection is Depending on Availability of Data



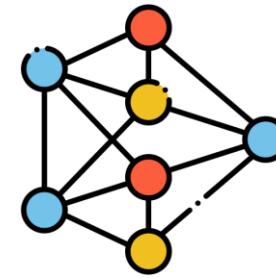
Machine Learning

- Is at the core of AI analytics, enabling models to learn from **historical data** and make **predictions about future outcomes**.



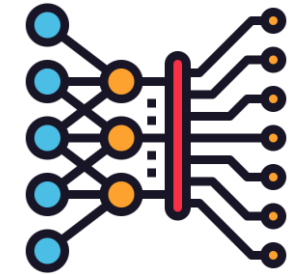
Natural language processing (NLP)

- It is essential for analyzing and interpreting human language, enabling AI tools to process and understand **text-based data**.



Neural Networks

- They represent a series of algorithms that **recognize relationships in data** and are particularly useful in **identifying complex patterns and correlations**.



Deep Learning

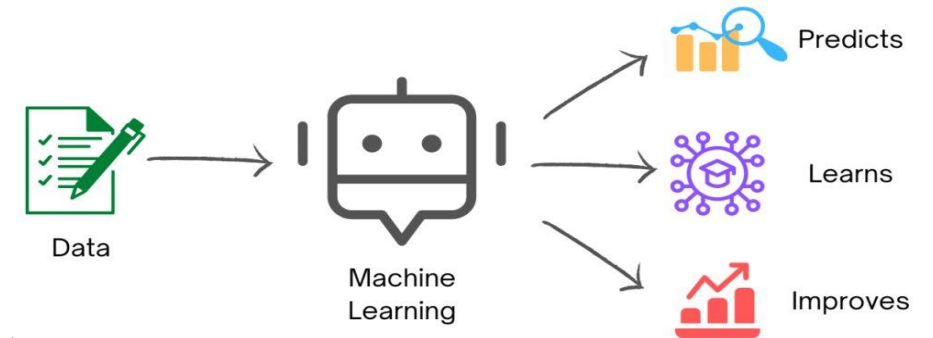
- Implies processing vast amounts of **unstructured data, such as images, videos, and text**, to generate highly detailed and nuanced insights.



4 Pillars of AI Analytics

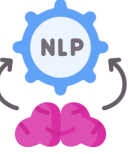
Machine Learning – A subset of AI that enables systems to learn from data and improve performance over time without explicit programming

Introduction to Machine Learning



Source: <https://doi.org/10.1007/978-3-319-20010-1>

Definition	Purpose	Key Features		Applications		Example Tools
A subset of AI that enables systems to learn from data and improve performance over time without explicit programming.	Identifying patterns, making predictions, and automating decision-making.	Uses algorithms like decision trees, support vector machines, and ensemble methods.	Focuses on supervised, unsupervised, and reinforcement learning.	Predictive maintenance in manufacturing	Fraud detection in banking. Customer segmentation in marketing.	Scikit-learn, TensorFlow, PyTorch.



4 Pillars of AI Analytics

Natural Language Processing (NLP) – The branch of AI focused on enabling computers to understand, interpret, and respond to human language



Source: https://doi.org/10.1007/978-81-322-3972-7_19

Definition

The branch of AI focused on enabling computers to understand, interpret, and respond to human language.

Purpose

Bridge communication between humans and machines.

Key Features

Analyzes text or speech data for sentiment, context, and meaning.

Techniques include tokenization, stemming, and named entity recognition.

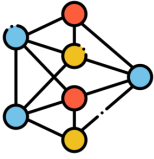
Applications

Chatbots and virtual assistants (e.g., Siri, Alexa).

Automated translation (e.g., Google Translate).

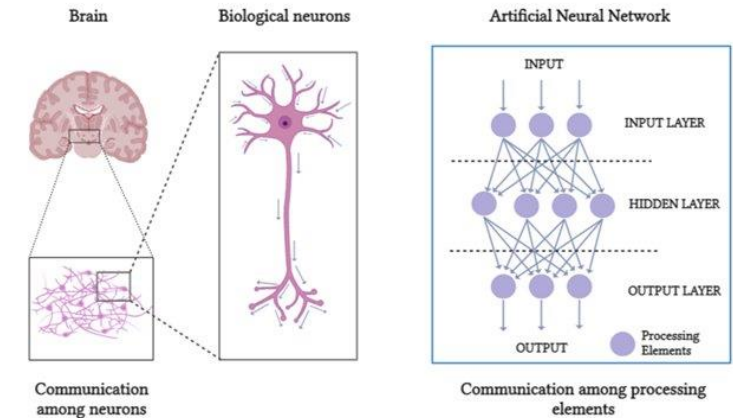
Example Tools

Chat-GPT



4 Pillars of AI Analytics

Neural Networks – Computing systems inspired by the human brain that are designed to recognize patterns and relationships in data



Source: <https://doi.org/10.1201/9781315273570>

Definition

Purpose

Key Features

Applications

Example
Tools

Computing systems inspired by the human brain that are designed to recognize patterns and relationships in data.

Model complex relationships in data and solve problems traditional algorithms struggle with.

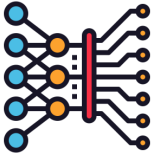
Composed of layers of interconnected nodes (neurons).

Includes feedforward, convolutional, and recurrent architectures.

Image recognition (e.g., facial recognition systems).

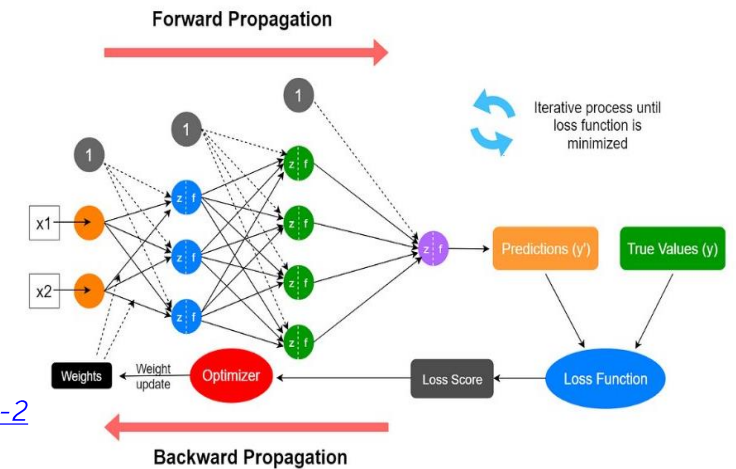
Pattern recognition in financial forecasting.

Keras, PyTorch, TensorFlow.



4 Pillars of AI Analytics

Deep Learning – A subset of machine learning that uses multi-layered neural networks to analyze vast amounts of data and extract high-level features



Source: <https://doi.org/10.1007/978-3-319-73004-2>

Definition	Purpose	Key Features	Applications		Example Tools	
A subset of machine learning that uses multi-layered neural networks to analyze vast amounts of data and extract high-level features.	Solve highly complex problems requiring hierarchical data representation.	Uses architectures like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs).	Requires large datasets and high computational power.	Autonomous vehicles (e.g., Tesla's self-driving cars).	Generative models (e.g., deepfake generation).	TensorFlow, PyTorch, OpenAI Codex.

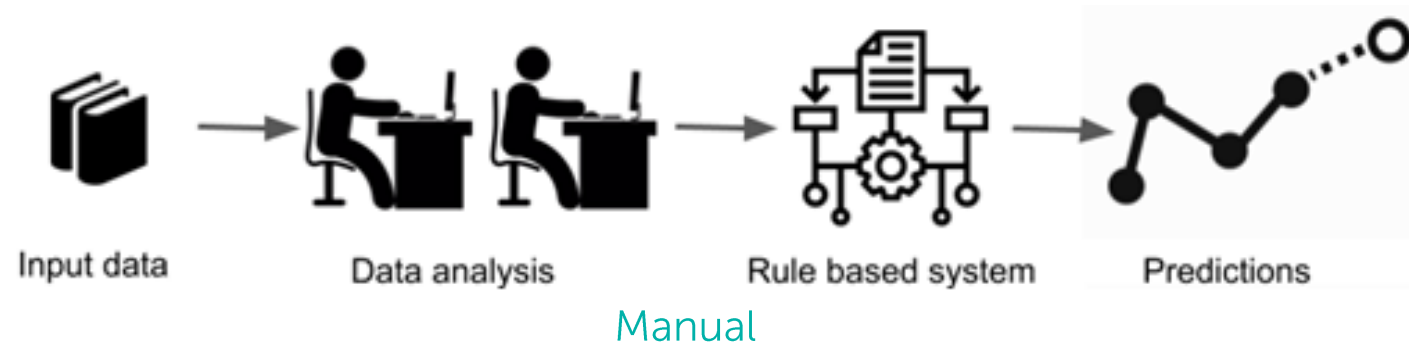
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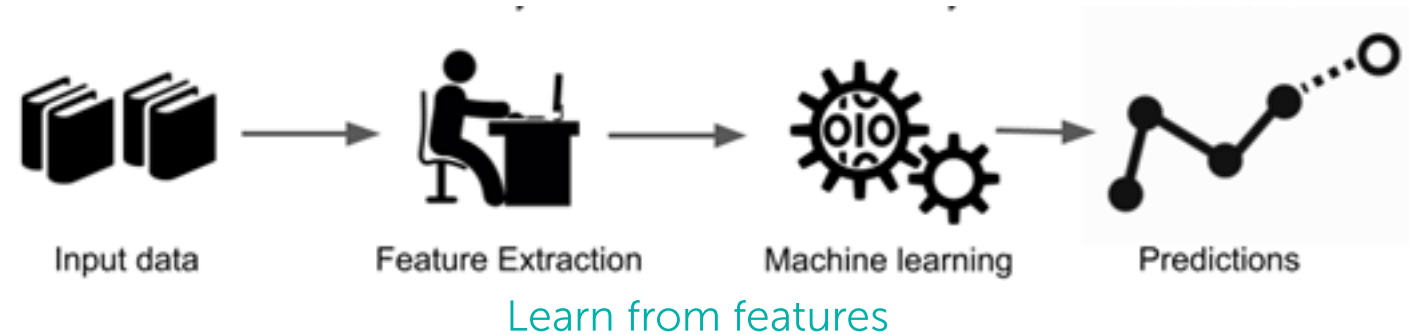
Evolution of AI Techniques

Rule-based, Machine Learning and Deep Learning

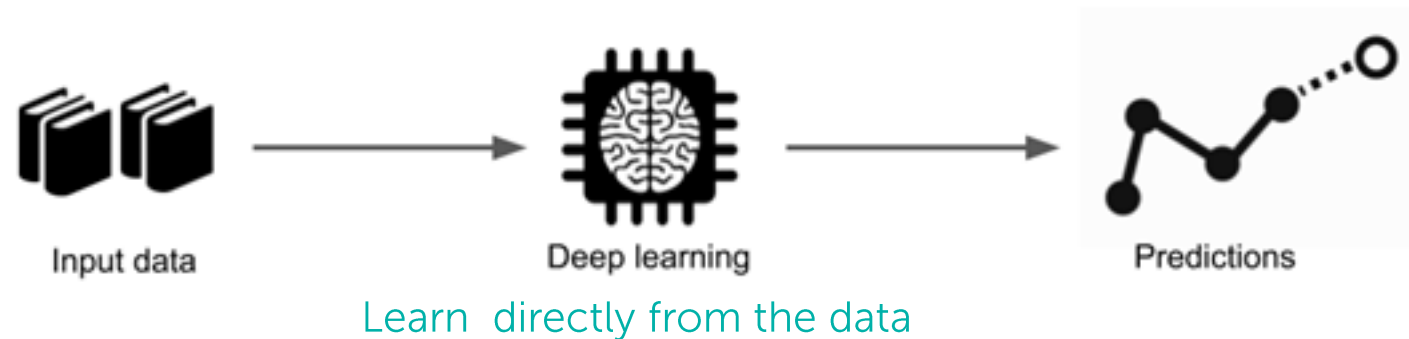
Rule-based Expert System



Machine Learning (ML)

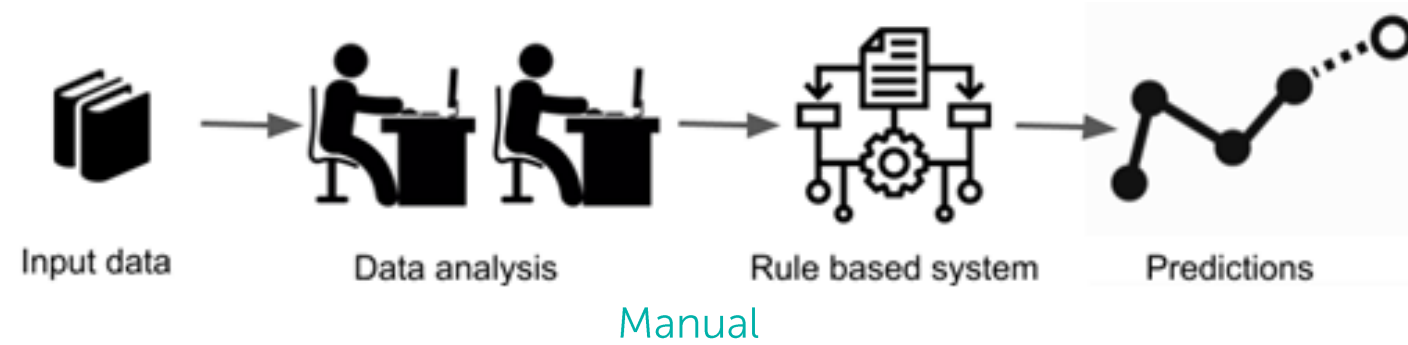


Deep Learning (DL)



Evolution of AI Techniques

Rule-based Technique



Timeframe

- 1950s–1980s

Characteristics

- Based on explicit rules crafted by domain experts.
- Uses **if-then** logic to make decisions or solve problems.
- Deterministic and interpretable.
- Highly dependent on expert knowledge.

Strengths

- Easy to understand and implement.
- Performs well in structured and well-defined domains.

Weaknesses

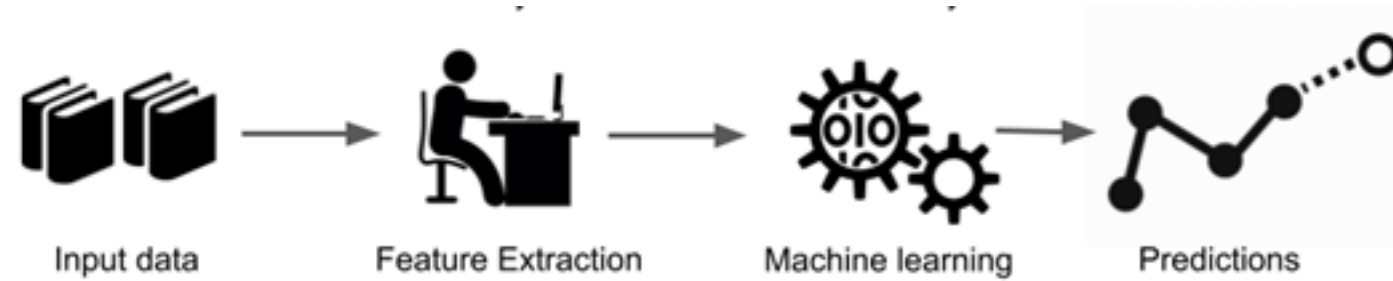
- Limited scalability for complex or dynamic environments.
- Difficult to maintain and update as knowledge grows.
- Cannot handle uncertainty or learn from data.

Applications

- Expert systems like MYCIN (medical diagnosis) and DENDRAL (chemical analysis).
- Early chatbots like ELIZA.

Evolution of AI Techniques

Machine Learning (ML) Technique

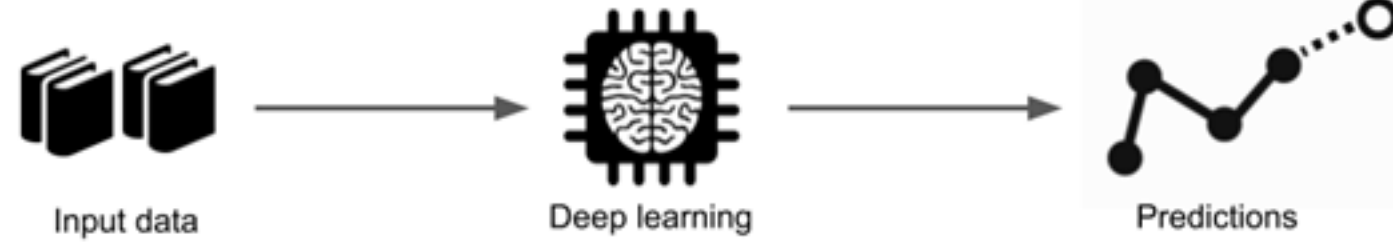


Learn from features

Timeframe	Characteristics	Strengths	Weaknesses	Applications
1980s–2010s	- Shifts from explicit programming to data-driven learning. - Uses algorithms to identify patterns and make predictions. - Involves techniques like supervised, unsupervised, and reinforcement learning.	- Can generalize from data, reducing reliance on explicit rules. - Better adaptability to real-world problems. - Handles a broader range of tasks, such as classification, regression, and clustering.	- Performance depends on the quality and quantity of labeled data. - Requires significant domain expertise for feature engineering. - Limited in processing unstructured data like images or raw text.	Spam detection, recommendation systems, fraud detection, and basic natural language processing (NLP).

Evolution of AI Techniques

Deep Learning (DL) Technique



Learn directly from the data

Timeframe	Characteristics	Strengths	Weaknesses	Applications
2010s–present	- A subset of ML focused on neural networks with multiple layers (deep neural networks). - Processes raw data and automatically extracts relevant features. - Handles unstructured data effectively (e.g., images, audio, text).	- Exceptional performance on complex tasks, such as image recognition, NLP, and speech recognition. - Scales well with large datasets and high-performance hardware. - Continuously improves with advances in architectures (e.g., GPT, BERT).	- High computational cost and energy consumption. - Requires extensive labeled data for supervised tasks. - Often opaque ("black-box") and challenging to interpret.	Autonomous vehicles, virtual assistants, advanced robotics, and creative AI (e.g., art and music generation).

Evolution of AI Techniques

Comparison Between Rule-based, ML and DL

Feature	Rule-Based Systems	ML	DL
Data Dependency	Minimal	Moderate	High
Feature Engineering	Minimal	Manual	Automatic
Scalability	Limited	Moderate	High
Interpretability	High	Moderate	Low
Performance on Complex Tasks	Poor	Moderate	High

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Standard Related to AI/ML

ISO/IEC Standards



ISO/IEC 22989:2022 - AI Concepts and Terminology

Provides a foundational understanding of AI concepts, terminology, and the scope of AI technologies.



ISO/IEC 23053:2022 - Framework for AI Systems Using Machine Learning

Offers guidance on the development of AI systems that incorporate machine learning components. Focuses on lifecycle management and performance evaluation.



ISO/IEC TR 24027:2021 - Bias in AI Systems and AI-Aided Decision Making

Addresses issues of bias in AI systems, providing strategies for identifying, managing, and mitigating bias.



ISO/IEC TR 24028:2020 - AI Trustworthiness

Covers trustworthiness aspects of AI, including transparency, explainability, and reliability.



ISO/IEC 38507:2022 - Governance of IT: Guidance for AI

Provides governance frameworks to oversee the ethical and effective deployment of AI systems.



ISO/IEC TR 24029 - Assessment of the Robustness of Neural Networks

Focuses on methods for evaluating the robustness of machine learning systems, particularly neural networks.



ISO/IEC 23894 - Risk Management in AI

(Under development) Aims to standardize risk assessment and mitigation strategies for AI systems.

Standard Related to AI/ML

IEEE Standards



IEEE 7000 Series - Ethical and Social Standards for AI

IEEE 7001: Transparency of Autonomous Systems.

IEEE 7002: Data Privacy Processes.

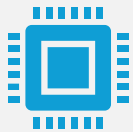
IEEE 7003: Algorithmic Bias Considerations.

IEEE 7005: Transparency in Machine Learning.



IEEE P2801 - Recommended Practice for the Quality Management of Datasets

Provides guidelines for the preparation and management of datasets used in training ML models.



IEEE P2811 - General Principles for AI

Establishes principles for designing AI systems with safety, accountability, and fairness in mind.

